

SM015
Mathematics 1
Trial Sem 1
Session 2025/2026
2 hours

SM015
Matematik 1
Percubaan Sem 1
Sesi 2025/2026
2 jam



KOLEJ MATRIKULASI MELAKA

PEPERIKSAAN PERCUBAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME TRIAL EXAMINATION

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO

NAME	:	_____
MATRIC NUMBER	:	_____
TUTORIAL LECTURER	:	_____

UNTUK KEGUNAAN PEMERIKSA SAHAJA

FOR EXAMINER'S USE ONLY

SECTION A		SECTION B	
QUESTION	MARKS	QUESTION	MARKS
1		1	
2		2	
3		3	
		4	
		5	
		6	
		7	
TOTAL		TOTAL	





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UNIT MATEMATIK
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MATHEMATICS I SM015

LIST OF MATHEMATICAL FORMULAE

SENARAI RUMUS MATEMATIK

Trigonometry Trigonometri $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$ $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$ $\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$ $\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$ $\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$ $\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$ $\sin 2A = 2 \sin A \cos A$ $\cos 2A = \cos^2 A - \sin^2 A$ $= 2 \cos^2 A - 1$ $= 1 - 2 \sin^2 A$ $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$ $\sin^2 A = \frac{1 - \cos 2A}{2}$ $\cos^2 A = \frac{1 + \cos 2A}{2}$	Differentiation <table style="width: 100%;"> <tr> <td>$f(x)$</td><td>$f'(x)$</td></tr> <tr> <td>$\sin x$</td><td>$\cos x$</td></tr> <tr> <td>$\cos x$</td><td>$-\sin x$</td></tr> <tr> <td>$\tan x$</td><td>$\sec^2 x$</td></tr> <tr> <td>$\cot x$</td><td>$-\operatorname{cosec}^2 x$</td></tr> <tr> <td>$\sec x$</td><td>$\sec x \tan x$</td></tr> <tr> <td>$\operatorname{cosec} x$</td><td>$-\operatorname{cosec} x \cot x$</td></tr> </table> Sum Rule $\frac{d}{dx}(u(x) + v(x)) = \frac{du}{dx} + \frac{dv}{dx}$ Product Rule $\frac{d}{dx}(u(x) \cdot v(x)) = v \frac{du}{dx} + u \frac{dv}{dx}$ Quotient Rule $\frac{d}{dx}\left(\frac{u(x)}{v(x)}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$ Chain Rule $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$ $\frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{dy}{dx}\right) \cdot \frac{dx}{dt}$	$f(x)$	$f'(x)$	$\sin x$	$\cos x$	$\cos x$	$-\sin x$	$\tan x$	$\sec^2 x$	$\cot x$	$-\operatorname{cosec}^2 x$	$\sec x$	$\sec x \tan x$	$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
$f(x)$	$f'(x)$														
$\sin x$	$\cos x$														
$\cos x$	$-\sin x$														
$\tan x$	$\sec^2 x$														
$\cot x$	$-\operatorname{cosec}^2 x$														
$\sec x$	$\sec x \tan x$														
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$														

Sphere
Sfera

$$V = \frac{4}{3} \pi r^3$$

$$S = 4\pi r^2$$

Right circular cone
Kon membulat tegak

$$V = \frac{1}{3} \pi r^2 h$$

$$S = \pi r^2 + \pi r h$$

Right circular cylinder
Silinder membulat tegak

$$V = \pi r^2 h$$

$$S = 2\pi r^2 + 2\pi r h$$





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SECTION A [25 MARKS]

This section consists of **3 questions**. Answer **all** questions.

1. Solve

(a) If $f(x) = x^2 + 3$. Find $\lim_{x \rightarrow 0} \left(\frac{f(x+5) - f(5)}{x} \right)$

[5 marks]

(b) $\lim_{x \rightarrow +\infty} (\sqrt{x^2 + 3} - x)$

[4 marks]

2. (a) Given that f and g are functions such that $f(x) = x^3 g(x^2)$, $g(1) = 5$ and $g'(1) = 3$.
Find the value of $f'(1)$.

[5 marks]

(b) Given $f(x) = \frac{3x^2 - 5}{x}$ and $g(x) = \ln \sqrt{f(x)}$, find $g'(x)$.

[4 marks]

3. Given the curve $y = x^3 - \frac{3}{2}x^2 - 18x + 1$. Use the first derivative test to determine the maximum and minimum points of y .

[7 marks]





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SECTION B [75 MARKS]

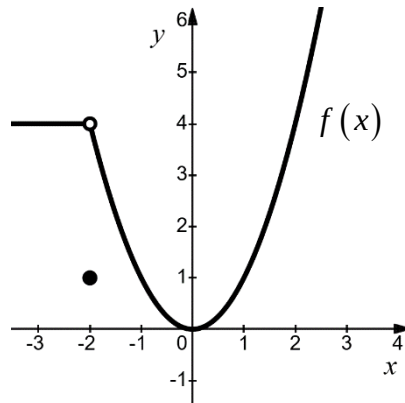
1. Given a complex number $z = \frac{(1+i)^2}{4-i}$.
- (a) Express z in the cartesian form $a+bi$, where a and b are real numbers.
[4 marks]
- (b) Find the argument of z correct to 3 significant numbers.
[2 marks]
2. Find
- (a) the value of x that satisfies the equation $\log_2 x = 2\log_4(x+4) - 1$.
[5 marks]
- (b) the solution set for the inequality $9x + \frac{25}{x} \geq 30$
[5 marks]
3. Given that $f(x) = \sqrt{x+4}$ where $x \geq -4$.
- (a) Show that the function is a one-to-one function algebraically.
[4 marks]
- (b) Find $f^{-1}(x)$.
[3 marks]
- (c) Sketch graphs of $f(x)$ and $f^{-1}(x)$ on the same coordinate axes.
[3 marks]
4. (a) If $g(x) = \frac{2x-3}{mx+2}$ where $m \neq 0$ and $x \neq -\frac{2}{m}$ such that $(g \circ g)(0) = \frac{12}{5}$.
Calculate the value of m .
[4 marks]
- (b) By finding the composite functions $(f \circ g)(x)$ and $(g \circ f)(x)$, determine if $f(x) = \frac{1}{3x+2}$ and $g(x) = \frac{1-2x}{3x}$ are inverse function of each other.
[6 marks]





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5. a)



From the graph of $f(x)$ above, determine ;

i) $\lim_{x \rightarrow -2} f(x)$, if exist. **[2marks]**

ii) whether $f(x)$ is continuous at $x = -2$. **[2marks]**

b) Given
$$f(x) = \begin{cases} x+1 & x < -1 \\ x^2 - 2x - m & -1 \leq x < 0 \\ \frac{3x+1}{\sqrt{x^2-1}} & x \geq 0 \end{cases}$$

i) Find the value of m given that $f(x)$ is continuous at $x = -1$. **[4marks]**

ii) Find the vertical and horizontal asymptote of $f(x)$ for $x \geq 0$, if exist. **[6marks]**





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6. a) If $y = \sin(x^2 + 3)$, show that $x \frac{d^2y}{dx^2} - \frac{dy}{dx} + 4x^3y = 0$. **[5 marks]**

b) The parametric equations of a curve are given by $x = \cot(t) - \operatorname{cosec}(t)$ and $y = \tan(t) - \sec(t)$ where $0 < t < \frac{1}{2}\pi$.

i) Show that $x - \frac{1}{x} = 2 \cot(t)$ and $y - \frac{1}{y} = 2 \tan(t)$.

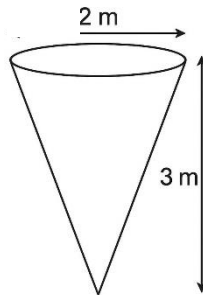
Deduce the value of $\left(x - \frac{1}{x}\right)\left(y - \frac{1}{y}\right)$. **[5marks]**

ii) Find $\frac{dy}{dx}$ and express your answer in terms of x and y only.

Hence, determine the gradient of the curve at the point $\left(-\frac{1}{2}, -\frac{1}{3}\right)$.

[6marks]

7. An inverted conical tank with radius 2m and height 3m is completely filled with water. Water is leaking from the vertex of the tank at a rate of 0.1m^3 per minute.



At the instant when the height of the water is 1m, calculate the rate of change of

a) the water level. **[7marks]**

b) the radius of the water surface. **[2marks]**

